

1. Given $(a, b) = (105, 1309)$ to find the $\gcd(105, 1309)$ using the Euclidean algorithm:
 - ① $1309/105 = 12$ with a remainder, $R = 49$
 - ② $105/49 = 2$ $R = 7$
 - ③ $49/7 = 7$ $R = 0$ $\therefore \gcd(105, 1309) = \gcd(49, 105) = \gcd(7, 49) = \gcd(0, 7) = 7$
2. Given $(a, b) = (105, 1309)$ to find the Bézout identities of (a, b) using the Euclidean algorithm solved above

$$\textcircled{1} \quad 7 = 105 - 2 \cdot 49$$

$$\textcircled{2} \quad 49 = 1309 - 12 \cdot 105$$

Substitute $\textcircled{2}$ into $\textcircled{1}$ yields:

$$7 = 105 - 2 \cdot (1309 - 12 \cdot 105)$$

$$7 = 105 - 2 \cdot 1309 + 24 \cdot 105$$

$$7 = 25 \cdot 105 - 2 \cdot 1309$$

$\therefore$ Bézout identities of (a, b) are 25 and -2 for 105 and 1309 respectively.

3. $m = 3677$ $a = 360$

To find the $\gcd$, start with the Euclidean algorithm:

$$\textcircled{1} \quad 3677/360 = 10 \quad R = 77$$

$$\textcircled{2} \quad 360/77 = 4 \quad R = 52$$

$$\textcircled{3} \quad 77/52 = 1 \quad R = 25$$

$$\textcircled{4} \quad 52/25 = 2 \quad R = 2$$

$$\textcircled{5} \quad 25/2 = 12 \quad R = 1$$

$$\textcircled{6} \quad 2/1 = 2 \quad R = 0$$

Since the last nonzero remainder is 1 then $\gcd(360, 3677) = 1$

$\therefore$ a is invertible modulo m because their $\gcd$ is one meaning that an inverse exists due to the two values being relatively prime towards one another with $m > 1$

Plugging $\textcircled{6}$ into $\textcircled{5}$ and $\textcircled{5}$ into $\textcircled{4}$ and so on...

$$1 = 173 \cdot 3677 - 1767 \cdot 360$$

$\therefore$ The inverse of a modulo m using Bézout identities of (m, a) are 173 and -1767 for 3677 and 360 respectively.